Classification

CSE 451
Advanced Topics in Computer Science
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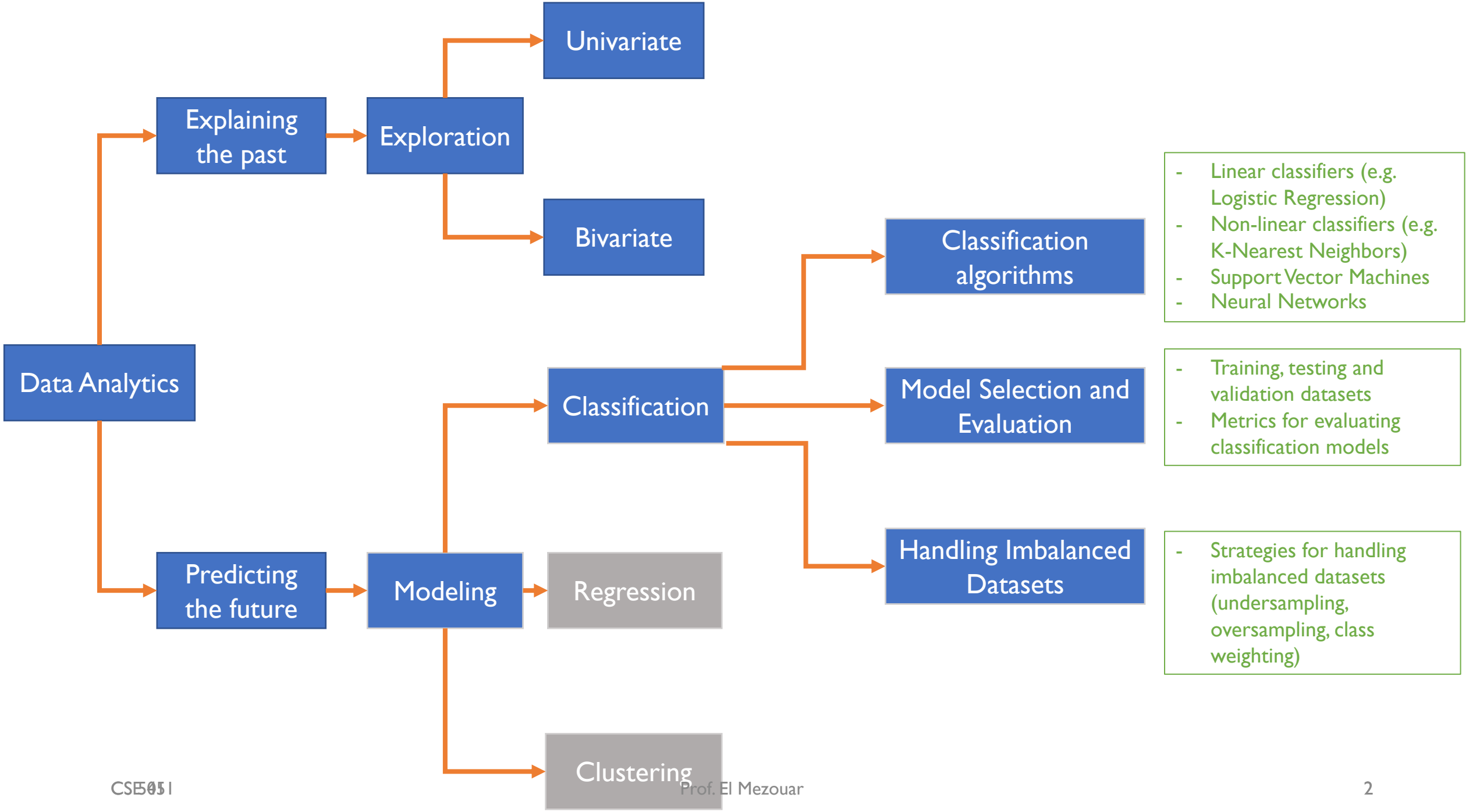

Data Analytics
Explaining the past
Exploration
Univariate
Bivariate
Predicting the future
Modeling
Classification
Regression
Clustering
Classification algorithms
Model Selection and Evaluation
Handling Imbalanced Datasets
- Linear classifiers (e.g. Logistic Regression)
- Non-linear classifiers (e.g. K-Nearest Neighbors)
- Support Vector Machines
- Neural Networks
- Training, testing and validation datasets
- Metrics for evaluating classification models
- Strategies for handling imbalanced datasets (undersampling, oversampling, class weighting)

Classical Machine Learning

Task Driven

Data Driven

Supervised Learning

(Pre Categorized Data)

- Classification
- Regression

Unsupervised Learning

(Unlabelled Data)

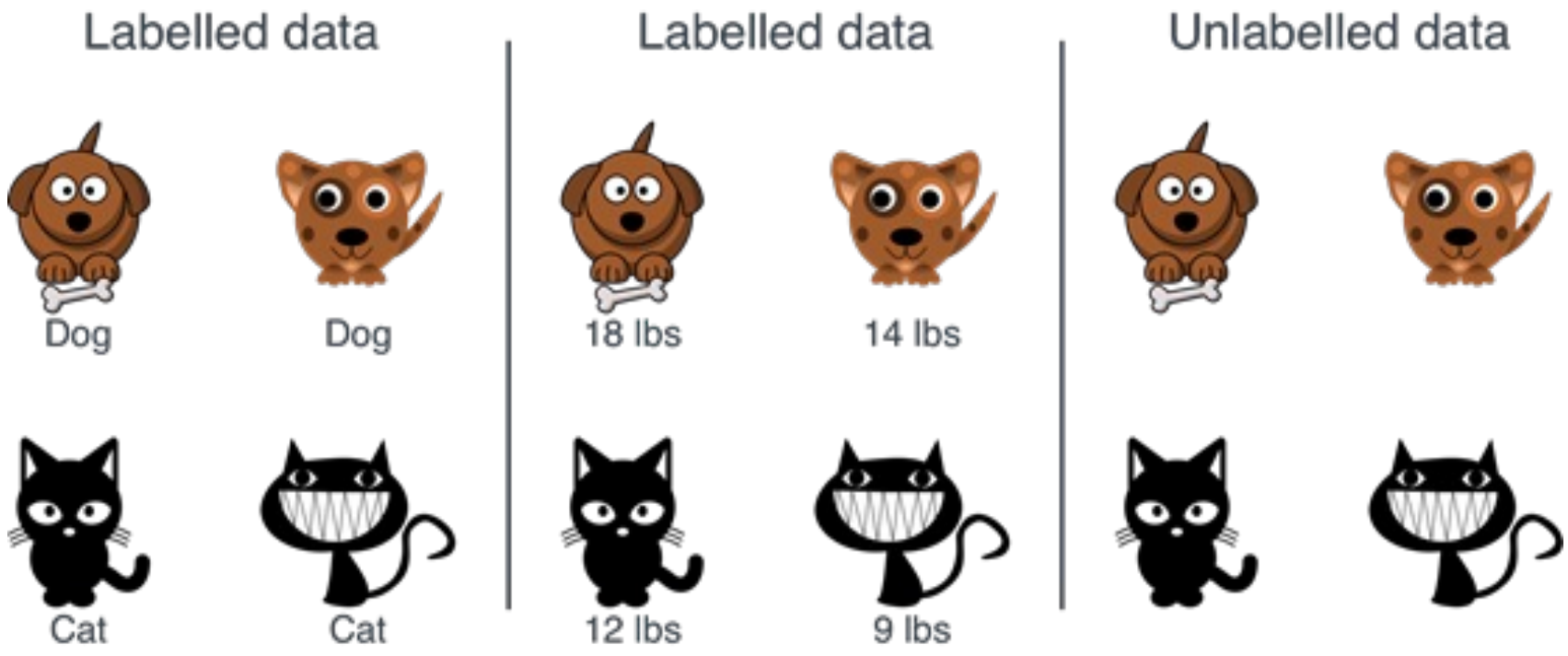
- Clustering
- Association
- Dimensionality Reduction

Output:

- Classification: A category
- Regression: A real-value
- Clustering: patterns within a group of uncategorized data
- Association: identify associations between different data objects
- Dimensionality Reduction: identify wider dependencies

Objective: Predictive analysis — Pattern recognition

Labeled vs Unlabeled data



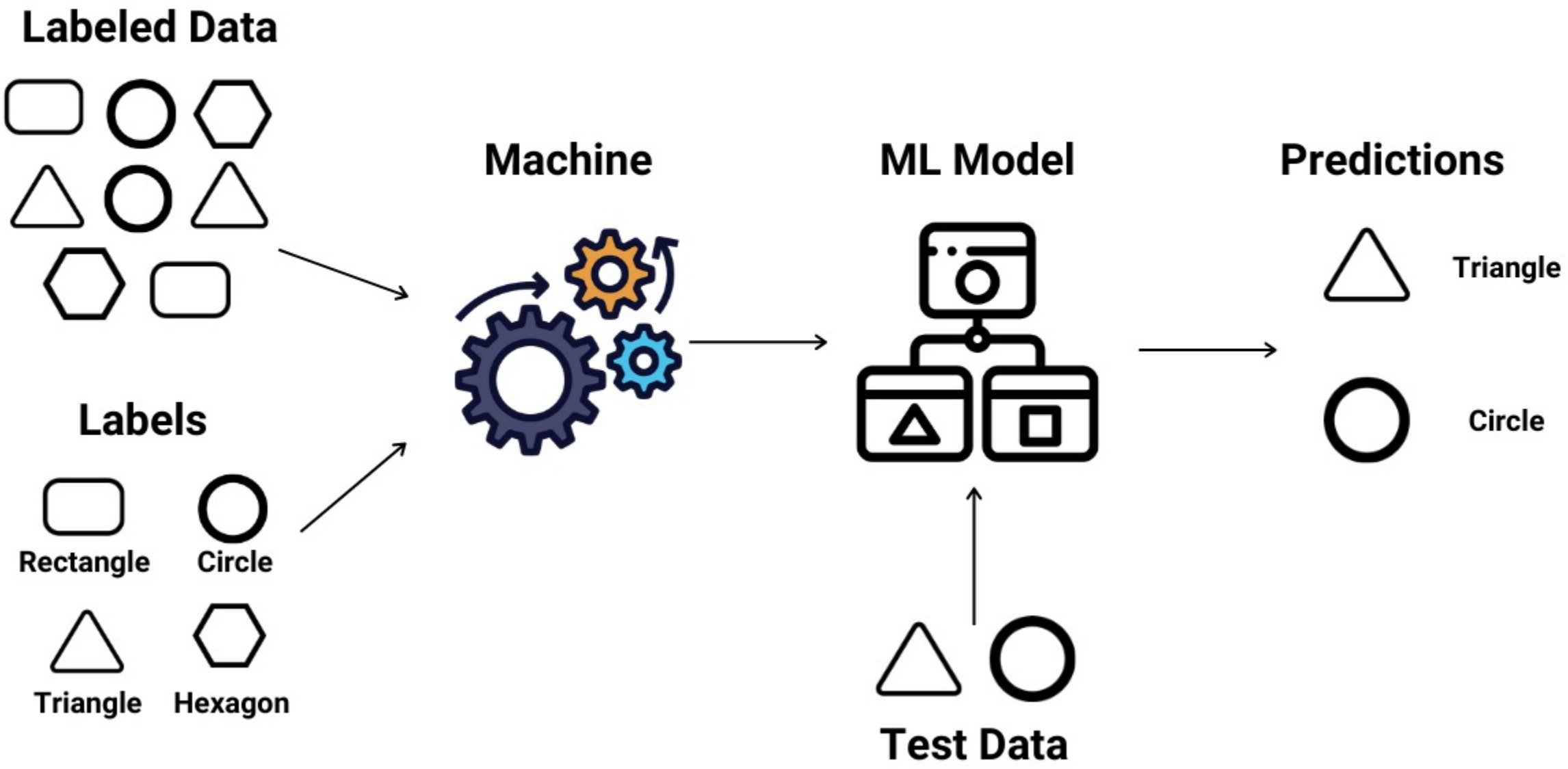
Introduction to Classification

What?

- Classification is a fundamental concept in the field of machine learning

- It involves identifying the category or class to which a new observation belongs based on a set of labeled training data.

- It is a supervised learning technique that is used to categorize or label a set of data into different classes or categories.


Labeled Data
Machine
ML Model
Predictions
Triangle
Circle
Labels
Rectangle
Circle
Triangle
Hexagon
Test Data

Types of classification

1) Binary Classification: predicting one of two possible outcomes, typically represented by 1 and 0, True and False, or Positive and Negative.
   - For example, classifying an email as spam or not spam, or diagnosing a patient as having a disease or not.
2) Multiclass Classification: predicting one of more than two possible outcomes.
   - For example, classifying an object as a car, bicycle, or motorcycle, or recognizing different types of fruits.
3) Multilabel Classification: predicting one or more outcomes for each sample. In other words, each sample can belong to multiple categories or classes at the same time.
   - For example, classifying a movie as belonging to multiple genres such as action, comedy, and drama.

Real-world applications of classification

- Image classification: recognizing objects or people in images and categorizing them into specific classes
- Spam filtering: classifying emails as spam or not spam
- Medical diagnosis: diagnosing diseases based on symptoms and test results
- Credit risk assessment: predicting the likelihood of a loan default based on various factors such as credit history, income, and job stability

Real-world applications of classification

- Sentiment analysis: classifying the sentiment of a piece of text as positive, negative, or neutral
- Customer segmentation: dividing customers into different groups based on their purchasing behavior and demographics
- Fraud detection: identifying fraudulent transactions in financial systems
- Marketing: classifying customers based on their likelihood to respond to a marketing campaign, or to purchase a certain product or service.

Basic Terminology

Feature and Target Variables

- Feature variables: (also called *predictors*, *inputs*, or *attributes*) are the variables used to describe an instance (such as an individual, item, or event).
  - These features are used to build a model that makes predictions about the target variable (also called response, label, or output).
- Target variable: is the variable that we want to predict based on the feature variables.
  - In a classification problem, the target variable is categorical (e.g. Yes/No, A/B/C), while in regression problems the target variable is continuous (e.g. age, salary, height).

Examples

- In a housing price prediction dataset, the feature variables could be the number of bedrooms, square footage, neighborhood, and so on, while the target variable would be the price of the house.

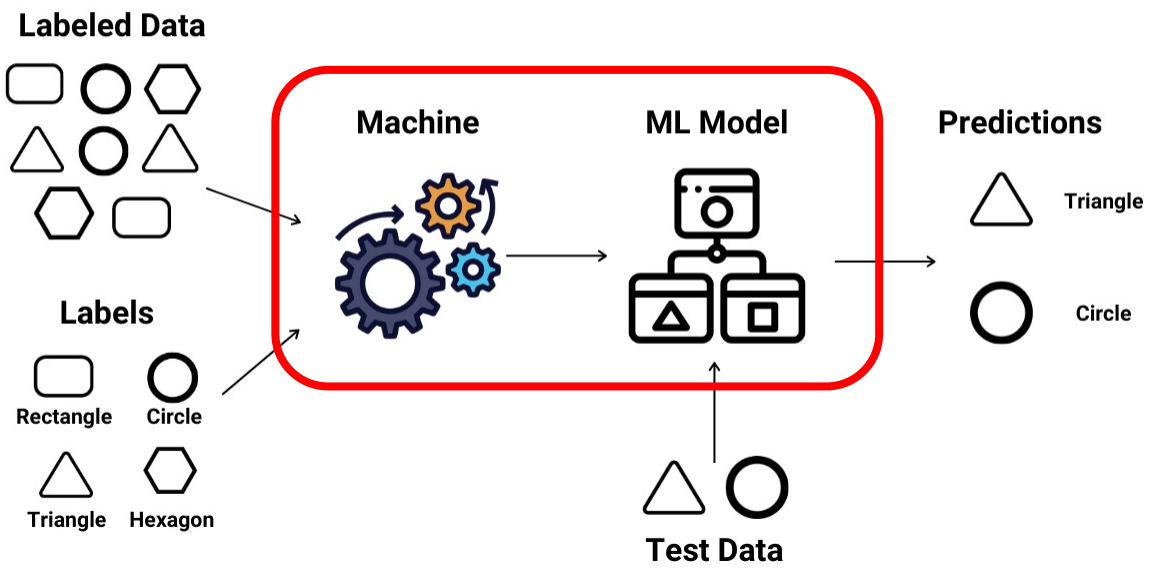
- In a medical diagnosis dataset, the feature variables could be patient symptoms, medical history, and test results, while the target variable would be the diagnosis (e.g. flu, pneumonia, etc.).

Model Training

This is the process of building a machine learning model using a training dataset.

The model is trained to learn the relationship between the features (input variables) and the target variable.

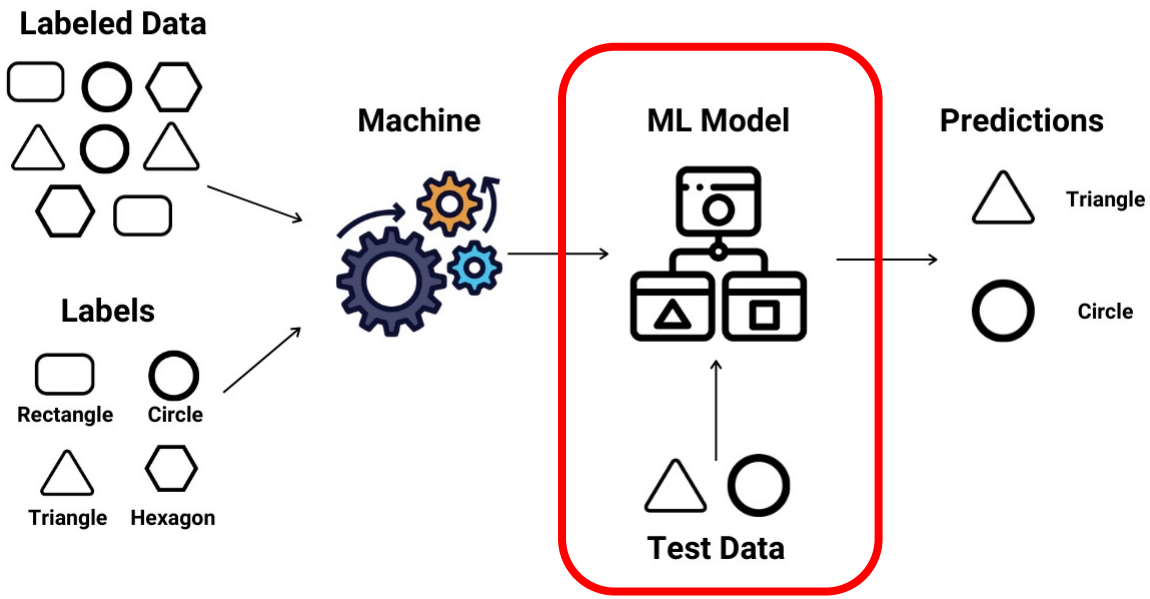
This process involves selecting an appropriate algorithm, defining the hyperparameters, and fitting the model to the training data.



Prediction

Once the model is trained, it can be used to make predictions on new, unseen data.

During prediction, the feature values are input into the model, and the target variable is predicted based on the learned relationship.



Overfitting and Underfitting

- Overfitting and underfitting are two common issues faced while training machine learning models.

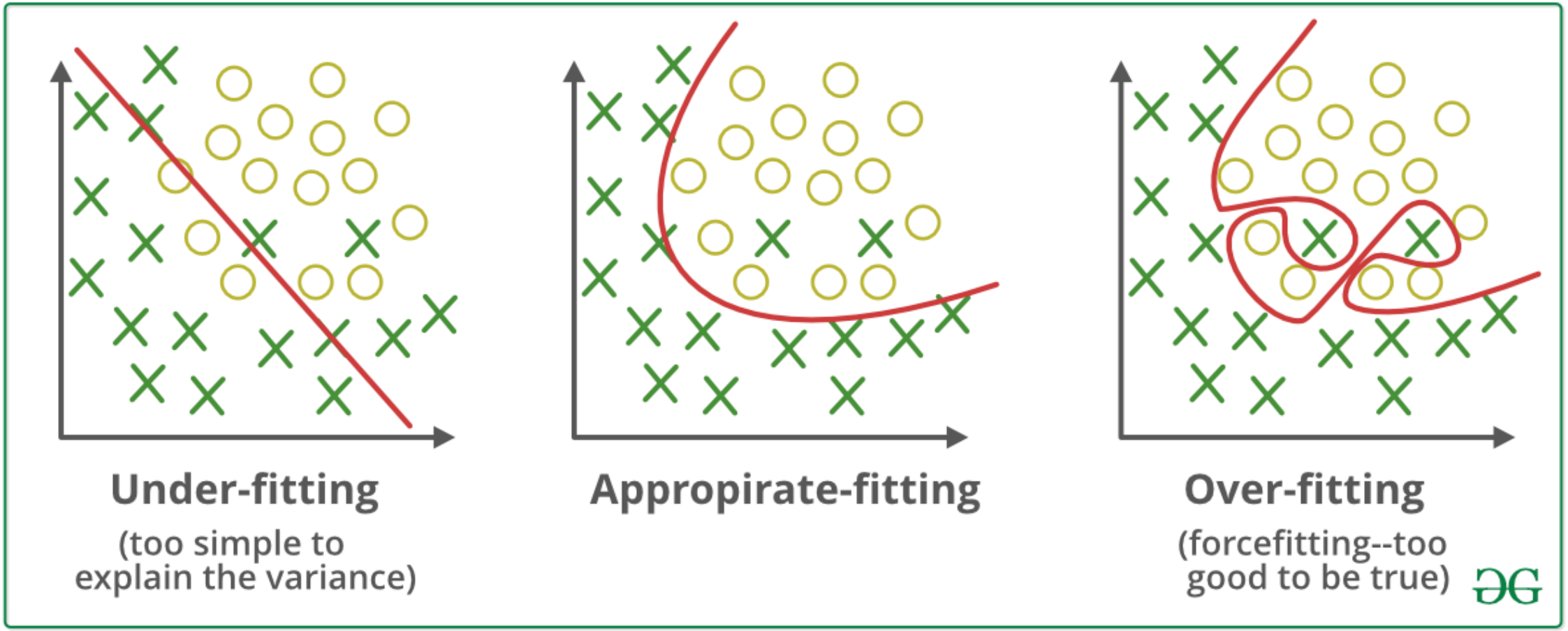
- Overfitting occurs when a model is trained too well on the training data and fits the noise in the data instead of the underlying pattern.
  - As a result, it performs well on the training data but poorly on the unseen data or validation data.
  - Overfitting can be identified by having a high accuracy on the training data but a low accuracy on the validation data.

Overfitting and Underfitting

- Underfitting, on the other hand, occurs when a model is not complex enough to capture the underlying pattern in the data. It results in a low accuracy on both the training and validation data.

It is important to strike a balance between overfitting and underfitting to build an effective model.

Overfitting and Underfitting



Bias and Variance

- Bias and variance are two important concepts in machine learning that describe the error in a model's predictions.

- Bias refers to the error that is introduced by assuming that the relationship between the features and target is too simple.
  - A model with high bias pays little attention to the training data and oversimplifies the relationship between the features and target.
  - As a result, it often has a high training error and a high test error.

Bias and Variance

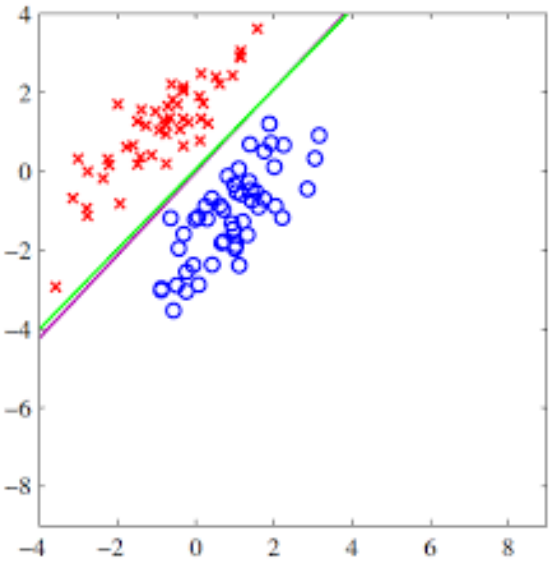
- Variance, on the other hand, refers to the error that is introduced by the model being too complex and fitting the training data too closely.
  - A model with high variance pays too much attention to the training data and overfits it, capturing the noise in the data as well as the underlying relationship.
  - As a result, it has a low training error but a high test error.

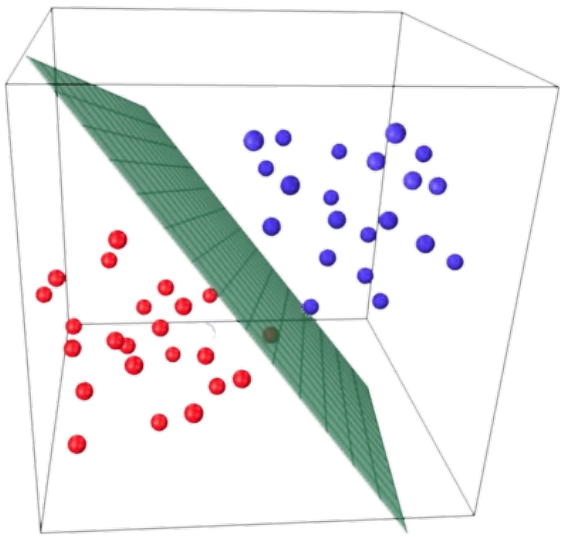
The goal in building a machine learning model is to find a balance between bias and variance to minimize the total error. This is often referred to as the bias-variance trade-off.

Classification Algorithms

Linear classifiers

- A linear classifier is a machine learning algorithm that uses a linear function to separate data into different classes.
- The goal of a linear classifier is to find the hyperplane (*a line* or *a plane* in high-dimensional space) that best separates the data into their respective classes.
- The hyperplane is defined by a set of coefficients that are estimated during the training phase.





Linear classifiers

Examples of linear classifiers include

- Logistic Regression,
- Support Vector Machines (SVM) with linear kernels, and
- Linear Discriminant Analysis (LDA).

These algorithms make predictions based on the values of the features and the coefficients of the hyperplane, which are used to determine the class of an observation.

Logistic Regression: Definition

- Logistic Regression is a popular supervised machine learning algorithm used for binary classification problems.

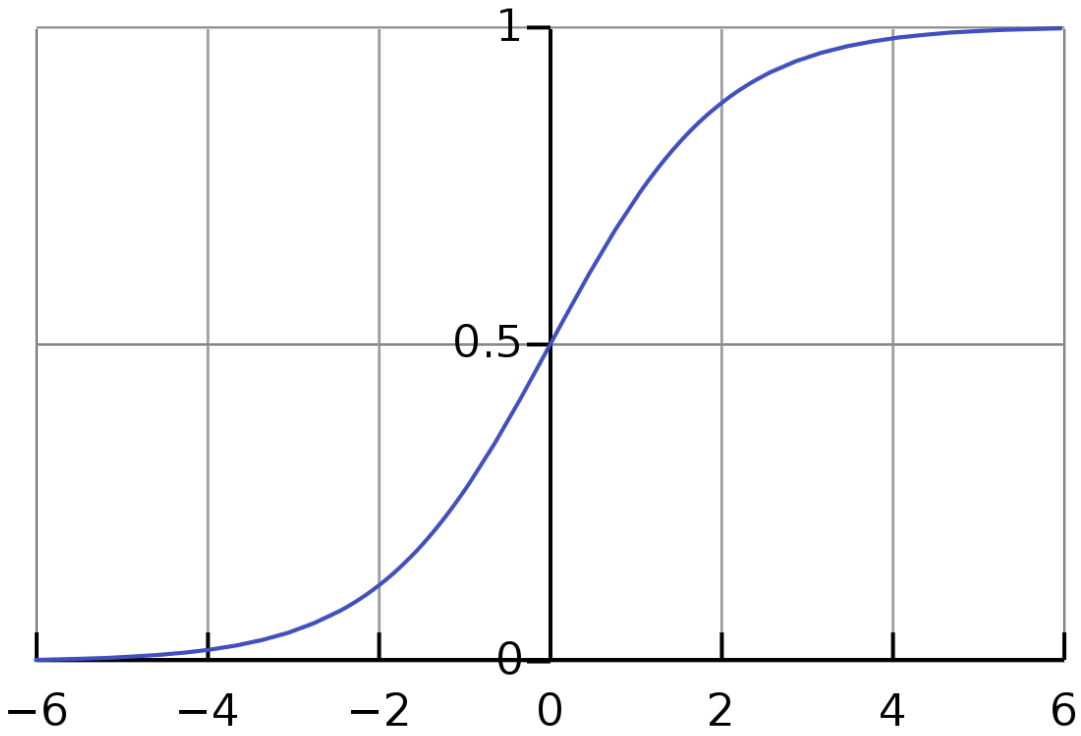
- In logistic regression, the target variable is binary and the prediction is made based on the relationship between the independent (or feature) variables and the dependent (or target) variable.

- The main objective of logistic regression is to find the best fitting model (i.e., a line or hyperplane) that separates the classes in the feature space.

Logistic Regression: How it works

- The algorithm works by modeling the probability of an event occurring (e.g., a customer buying a product) using a sigmoid function (the logistic function).

- The output of the logistic regression model is a probability score between 0 and 1, which can then be used to make a binary classification.



$$f(x) = \frac{1}{1 + e^{-x}}$$

Logistic Regression: How it works

$$f(x) = \frac{1}{1 + e^{-x}}$$

- x is a linear combination of one or more features in the dataset.
- f(x) is a probability between 0 and 1.
  - For example, if the output of the function is above 0.5, the output is considered as 1. On the other hand, if the output is less than 0.5, the output is classified as 0.

Model Selection and Evaluation

Training, Training and Validation datasets

Training, testing, and validation datasets are used in the process of developing and evaluating a machine learning model.

1. Training dataset: This dataset is used to train the machine learning model.
   - The model is trained by fitting the model to the training data. The model learns the patterns in the data and uses them to make predictions.

Training, Training and Validation datasets

2. Testing dataset: This dataset is used to evaluate the performance of the machine learning model after it has been trained.

- The model is presented with new, unseen data and it makes predictions based on what it has learned from the training data.
- The accuracy of these predictions is then used to evaluate the performance of the model.

Testing, Training and Validation datasets

3. Validation dataset: This dataset is used to tune the hyperparameters of the machine learning model.

- The model is trained on the training data and then evaluated on the validation data.
- The hyperparameters are adjusted based on the performance of the model on the validation data.
- This helps to prevent overfitting of the model to the training data.

Metrics for evaluating classification models

Classification models can be evaluated using a variety of metrics, depending on the specific use case and requirements.

Some of the most used metrics are:

1. Accuracy
2. Confusion matrix
3. Precision
4. Recall
5. F1 score
6. ROC curve (Receiver Operating Characteristic)
7. AUC (Area Under the Curve)

Accuracy

Ratio of correct predictions made by a classifier to the total number of predictions made

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}}$$

Confusion Matrix

A 2-D table that shows the number of true positive, true negative, false positive, and false negative predictions made by the model.

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

The entries in the matrix can then be used to calculate various performance metrics, such as precision, recall, F1-score, and AUC for the ROC curve.

Precision

the number of true positive predictions (i.e. positive predictions that are actually correct) divided by the total number of positive predictions made by the model.

$$\text{Precision} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Positives}}$$

Recall

The proportion of actual positive instances that are correctly classified as positive by the model.

Also called TPR (True Positive Rate) or Sensitivity.

$$\text{Recall} = \frac{\text{True Positives}}{\text{True Positives} + \text{False Negatives}}$$

F1 score

A metric that combines precision and recall. It is calculated as the harmonic mean of precision and recall.

$$\text{F1-score} = \frac{2 * (\text{Precision} * \text{Recall})}{\text{Precision} + \text{Recall}}$$

The F1 Score ranges between 0 and 1, with 1 being the best possible score and 0 the worst.

ROC curve

- A graphical representation of the performance of a binary classification model as the *discrimination threshold (probability threshold)* is varied.

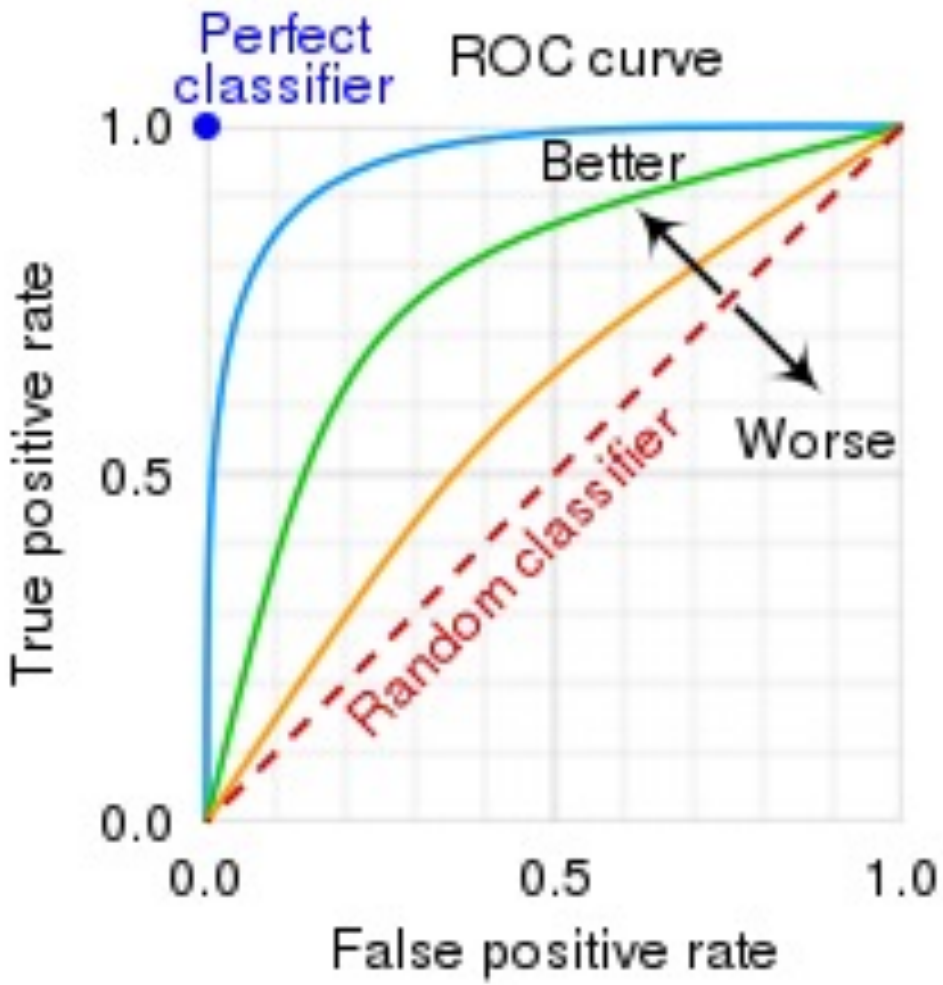
- It plots the true positive rate (TPR) against the false positive rate (FPR) at various threshold settings.

- The ROC curve is a useful tool for evaluating the trade-off between the true positive rate and the false positive rate of a classifier.

ROC curve

- FPR = FP / (FP + TN)
  i.e. probability of false alarm

- TPR = TP / (TP + FN)
  i.e. probability of detection



AUC

- The interpretation of the ROC curve is based on the Area Under the Curve (AUC), which summarizes the overall performance of the model.

- An AUC of 1 indicates a perfect model, while an AUC of 0.5 represents a random model.

- A higher AUC value indicates a better performance, with a larger area under the curve meaning a greater balance between TPR and FPR.

Handling Imbalanced Datasets

What is an imbalanced dataset?

- A type of dataset where the distribution of classes is not equal or balanced.
  - In other words, one class has significantly more or fewer observations than the other class(es).

- Quite common in real-life scenarios, such as credit card fraud detection, disease diagnosis, or customer churn prediction.
  - Which class do we expect to be dominant in each example?

- For example, in fraud detection, fraudulent transactions may be rare and account for less than 1% of the total transactions.

Impact of an imbalanced dataset

Imbalanced dataset ➔ Biased Model

- The model may perform well on the majority class but may perform poorly on the minority class (high false-negative rate).
- Can be costly in certain applications such as fraud detection, and rare disease detection.

Strategies for handling imbalanced datasets

Resampling

either oversampling the minority class or undersampling the majority class to create a more balanced dataset.

- Oversampling techniques include *duplication*, *bootstrapping*, and *SMOTE*.
- Undersampling techniques include *random undersampling* and *Tomek links*.

Strategies for handling imbalanced datasets

Resampling: oversampling

Duplication: creating additional copies of the minority class instances in the dataset to balance the class distribution

Bootstrapping: a resampling technique that involves randomly sampling the original dataset with replacement to create multiple new datasets.

- These new datasets have the same number of instances as the original dataset but may have some instances *repeated* and some *omitted*.

Strategies for handling imbalanced datasets

Resampling: oversampling

SMOTE: Synthetic Minority Over-sampling Technique is a technique for generating synthetic samples for the minority class in imbalanced datasets.

Strategies for handling imbalanced datasets

Resampling: undersampling

Random undersampling: is a technique that involves randomly removing instances from the majority class to balance the class distribution.

Tomek links are pairs of instances, one from the minority class and one from the majority class, that are closest to each other. Removing the majority instance from each Tomek link can improve the separation between the two classes.

This technique is called Tomek links undersampling.

Strategies for handling imbalanced datasets

Class weighing

Class weighting is a technique used to handle imbalanced datasets by adjusting the weight of each class during model training.

The idea behind class weighting is to give more importance to the minority class by increasing its weight.

This approach can help the model to learn more from the minority class and produce better predictions for the imbalanced dataset.

Strategies for handling imbalanced datasets

Class weighing: example with logistic regression

```
from sklearn.linear_model import LogisticRegression

# Define class weights
class_weights = {0: 0.2, 1: 0.8}

# Create logistic regression object with class weights
lr = LogisticRegression(class_weight=class_weights)

# Train the model using the training data
lr.fit(X_train, y_train)
```

Demo: Logistic Regression